vector cheatsheet

Basic properties

$\vec{OP} = xi + yj + zk = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

notation

a, $\underset{\sim}{a}$ vector

$\hat{a}$ unit vector in direction of a

$|a|$ magnitude

$\hat{a} = \frac{\underset{\sim}{a}}{|a|}$

Equal vectors

- same magnitude
- same direction

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$

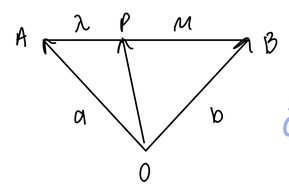
magnitude

$|\vec{OP}| = \left|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right| = \sqrt{x^2+y^2+z^2}$

$|a \times b|^2 = |a|^2|b|^2 - 2(a \cdot b)^2$

* can't square vectors, only their magnitude

ratio theorem



$\vec{OP} = \frac{\mu a + \lambda b}{\lambda + \mu}$

relationship between 2 vectors

1 (scalar) dot product

* results in a constant

How much 2 vectors are aligned
(component of one vector onto the other)

algebraic definition

$a \cdot b = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$

$= a_1b_1 + a_2b_2 + a_3b_3$

geometric definition

$\cos\theta = \frac{a \cdot b}{|a||b|}$



geometric definition

cosine (trigo identity)

proof:

$|a-b| = |a|^2 + |b|^2 - 2|a||b|\cos\theta$

$(a_1-b_1)^2 + (a_2-b_2)^2 + (a_3-b_3)^2$

$= a_1b_1 + a_2b_2 + a_3b_3$

$= |a||b|\cos\theta$

$\cos\theta < 0$	$\cos\theta = 0$	$\cos\theta > 0$
obtuse	90°	acute

Laws

commutative law	$a \cdot b = b \cdot a$
Associative law	$a \cdot (\lambda b) = \lambda(a \cdot b) = (\lambda a) \cdot b$
distributive laws	$a \cdot (b+c) = a \cdot b + a \cdot c$ $(b+c) \cdot a = b \cdot a + c \cdot a$
magnitude	$|a|^2 = a \cdot a$

dot product of vector onto itself

$a \cdot a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$

$= a_1^2 + a_2^2 + a_3^2$

$= |a|^2$

if $a \cdot b = 0$

$a = 0$

$b = 0$

$\perp$ b (as $\cos 90° = 0$)

↓

(projection)

2 (vector) cross product

* results in a vector

- $a \times b \rightarrow a$ curl to b
  $b \times a \rightarrow b$ curl to a
- always ⊥

$a \times b = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$

$= \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$

$= \underbrace{(|a||b|\sin\theta)}_{\text{magnitude}}\underbrace{\hat{n}}_{\text{direction}}$

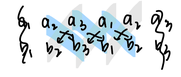
$\hat{n}$ is a unit vector ⊥ to a and b

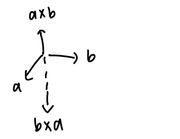
$a \times a = 0$

$\sin 0 = 0$

$a \times b = 0$ { $a = 0$, $b = 0$, $a \parallel b$

a 0 vector is produced $a \times b$ if $a \parallel b$





$a \times b \rightarrow$ curl $a \rightarrow b$
$b \times a \rightarrow$ curl $b \rightarrow a$

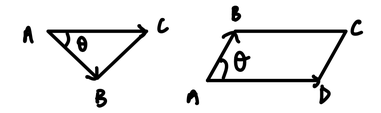
Area of triangle between 3 vectors

1) Form 2 vectors with common point $\vec{A}$

$\vec{u} = \vec{B} - \vec{A}$

$\vec{v} = \vec{C} - \vec{A}$

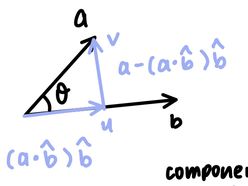
then use $\frac{1}{2}|\vec{u} \times \vec{v}|$

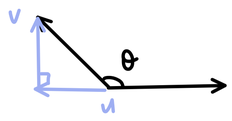


Area $= \frac{1}{2}|\vec{AB} \times \vec{AC}|$

Area $= |\vec{AB} \times \vec{AD}|$

resolving a vector into different components





component of a on b

vectors

$u \parallel b$

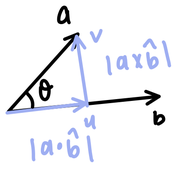
$u + v = a$

$v \perp u \perp b$

u is a projection of a onto b $(a \cdot \hat{b})\hat{b} = \frac{(\vec{a} \cdot \vec{b})}{|b|^2}\vec{b}$ $\frac{|\vec{a} \cdot \vec{b}|}{|\vec{b}|}$

v is a vector component of $a \perp b$ $a - (a \cdot \hat{b})\hat{b} = |a \times \hat{b}|$

magnitude

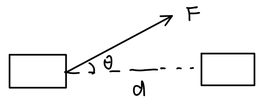


$|a \times \hat{b}| \rightarrow |a \times \hat{b}|$

$= \sqrt{|a|^2 - |a \cdot b|^2}$

$= |a - (a \cdot \vec{b})\vec{b}|$

application



$W_d = |F|\cos\theta\,|d|$

$= F \cdot d$